

Surface Engineering for Look and Feel

Improving the look and feel of single-use wooden cutlery
by surface engineering

Desiree Kasper and Victor Ploeger

University of Twente
16-04-2026

Abstract

The European Union (EU) created a goal to reduce packaging waste by 15% by 2040. Therefore, it is necessary to decrease plastics, such as plastic single-use cutlery. One alternative is single-use wooden cutlery, which has seen widespread adoption over the past few years and is expected to grow more.

A problem regarding these newly introduced single-use wooden cutlery is the lack of sensory research performed on them. The perception of the food can be influenced by the sensory aspects of the cutlery that is used.

Issues found in literature are separated into three main categories: tactile, aesthetic and flavour. Key findings are an atypical woody flavour, high roughness and low hardness.

Possible solutions to the key findings are: surface treatments, coatings and textures. The effect of these solutions will have to be tested on single-use wooden cutlery to verify effectiveness.

A test plan is set up to test the effectiveness of the found solutions, including an objective part on the mechanical and physical parameters and a subjective part on the effect on the user experience.

Contents

1	Introduction	3
2	Methodology	4
3	Research Questions	4
4	Problem definition	5
4.1	Tactile	5
4.2	Aesthetic	5
4.3	Flavour	7
4.4	Findings	7
5	Possible Solutions	8
5.1	Current manufacturing	8
5.2	Surface Treatments	8
5.2.1	CNC	8
5.2.2	High-speed friction	9
5.2.3	Thermomechanical densification	10
5.3	Ultrasonic densification	10
5.3.1	Findings	11
5.4	Coatings	11
5.4.1	Edible Coatings	11
5.4.2	Film Coating	11
5.4.3	Findings	11
5.5	Textures	11
5.6	Findings	13
6	Test plan	15
6.1	Objective tests	15
6.2	Subjective tests	15
7	Conclusion	16
8	Discussion	17
9	Acknowledgements	17

1 Introduction

Food consumption is a form of cultural expression, not just the meal itself, but also the context within which it is consumed. [24] Cutlery can be seen as one aspect of the context surrounding food, the only one to be haptically, visually, and orally perceived by the user.

Single-use cutlery is often used in the food delivery service, which has seen a rise in recent decades. Even before the COVID-19 pandemic, a significant increase in food orders was seen. [20] However, the COVID-19 pandemic also required more food safety, aiding even more in an increase of single-use plastics in the food industry. [21]

Plastic waste is known to be non-biodegradable, which has major environmental consequences. [22] The European Union (EU) created a goal to reduce packaging waste by 15% by 2040 [23]. Therefore, it is necessary to decrease single-use plastics.

Alternatives to single-use plastic tableware have been introduced and widely adopted in the form of biodegradable and reusable alternatives [15]. One such alternative is wooden cutlery, which has seen widespread adoption over the past years and is expected to grow more.[1]

Reusable cutlery is more environmentally friendly than single-use cutlery. However, comparing plastic, wooden and biodegradable cutlery, wooden cutlery has the smallest environmental impact. It has less impact on most of the environmental aspects, such as smog and global warming. [8]

A research performed on the life cycle assessment of single-use products explains that the disposal process of wooden cutlery is the most impactful process. Transportation is the second most impactful process. The production process on the other hand has little impact, so this process is the more environmentally sustainable process. [8]

Although wooden spoons have been dated all the way back to the sub-beolithic and bronze age [13], wood has not been the standard for cutlery in recent history. A problem regarding the reintroduced wooden cutlery is the lack of sensory research performed on them. The perception of the food can be influenced by the sensory aspects of the cutlery that is used [25]. Research on the emotional feelings about the different materials of cutlery shows that wooden tableware scored significantly lower on the ratings for ‘taste’ than regular and plastic tableware for most foods. This is most likely because wood introduces an atypical flavor, according to the article. [25] The participants acknowledged that, for example, with cold-served meals, they had negative emotions such as ‘unsatisfied’, ‘unpleasant’, and ‘irritated’ after eating these meals with wooden cutlery. [25]

This research aims to determine what factors impact the look and feel of wooden cutlery, and how these factors can be improved using surface engineering.

2 Methodology

First, this research aims to identify subjective and objective factors that influence perceptions of single-use wooden cutlery. A literature review will be performed on the subject and closely related subjects, such as the feel of wood in general. Afterwards, these sensory factors are translated into mechanical and physical parameters where possible.

Once an understanding of the issues users face when interacting with single-use wooden cutlery has been developed and translated into mechanical and physical parameters a literature review is conducted on possible ways to improve these parameters.

To test the possible solutions found in the literature on single-use wooden cutlery, a test plan is set up. Part of which objectively tests the mechanical and physical parameters, and part of it focuses on the subjective aspects to make sure the scientific conclusions translate to the user experience.

To perform this research, a main research question is formulated with a set of subquestions, which are found below.

3 Research Questions

The main research question has been formulated as: *“How can the look and feel of single-use wooden cutlery be improved while maintaining its mechanical, food-safety, and sustainability properties?”*

In order to answer this research question, the following subquestions have been formulated.

- What issues are found in single-use wooden cutlery?
- What, if any, surface finishes can tackle issues found in single-use wooden cutlery while being food-safe and maintaining mechanical and sustainability properties?
- What, if any, coatings can tackle issues found in single-use wooden cutlery while being food-safe and maintaining mechanical and sustainability properties?
- What, if any, (micro)textures can tackle issues found in single-use wooden cutlery?

4 Problem definition

Multiple aspects of cutlery are adjusted by changing its material, such as colour and weight. These aspects are also perceived through multiple senses; for example, changing the shape can influence both mouthfeel and emotional response through aesthetics. These issues were prevalent throughout most of the literature found on cutlery, but for this research, the selection process only included (parts of) literature with singled-out variables. The problem definition is divided into three sensory categories: tactile, aesthetic, and flavour.

4.1 Tactile

The tactile section is defined as all factors sensed through touch, whether oral or by hand.

In a survey, participants were asked to rank cutlery made of different materials. The materials they were asked to rank were: hard plastic, soft plastic, wood and paper. As a baseline measurement, a set of stainless steel cutlery was used. From these materials, hard plastic was the most acceptable material and paper the least. Wood was therefore not the most appealing, but not worse than paper. [25]

Another important aspect that became clear from the experiment was that the wooden cutlery had a significant surface roughness and heaviness. Another noticeable feature that the participants pointed out was that the teeth of the forks were not as sharp as preferred. [25]

A material change will result in a difference in strength and weight, among other material properties. In one research, fifty participants were asked to eat from two identical bowls of yoghurt, once using a stainless steel spoon and once using a replica made of metallic-plastic. The quality of the yoghurt, the overall experience and the taste were deemed significantly better when eating with the stainless steel spoon compared to the metallic-plastic spoon. [27] Weight can be defined by material and shape, and is considered a separate factor in another study. According to this research, spoon weight affected the food's taste, density, price estimation, and overall perception. [11]

A higher roughness in the dining environment is associated with a relaxed, warm atmosphere, while smooth materials are associated with cleanliness and modernity. Another association made in this research linked a higher hardness to a more functional, formal and reliable setting, while a lower hardness was found to be related to a casual and relaxed setting. The research also found another association with hardness, claiming that a higher hardness would enhance taste associations. [10]

4.2 Aesthetic

Mapping colour to taste is a concept that has been described in various literature; this research looked into the effect of five differently coloured spoons on various aspects of yoghurt taste. Only significant differences were found in the saltiness of the experienced food, but trends were identified when looking at the data gathered regarding the perceived sweetness, expensiveness and overall liking. [11]



Figure 1: Spoon colour variation

This research suggests the match between color and taste might be more primal than was previously assumed. Across cultures and the past three decades, the same links were seen between the flavours bitter, sweet, salty, sour and umami and certain colors. While different cultures use these colors differently in their cuisine repertoire. [28]

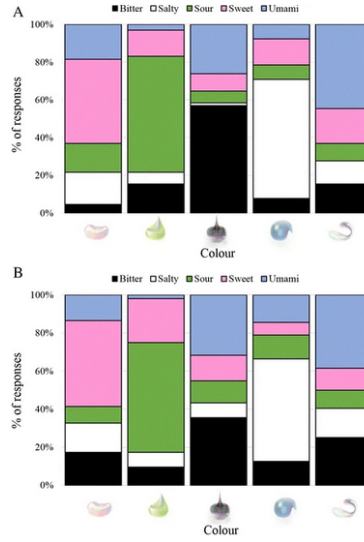


Figure 2: Colour matched to taste [28]

Shape can also play an important factor in the perceived taste of food. One research study, including thirty participants between the ages of twenty and thirty years, served yoghurt in a variety of bowls and spoons. The three types of spoons mainly differed in size and shape. Participants were asked to fill out a seven-point Likert scale on their first impression, color, smell, density, taste and the general impression. Even though very little can be said about the way a spoon for yoghurt should look, this research showed that the mean perception of yoghurt can be impacted by spoons of similar color and material, but varying sizes and shapes. [29]

4.3 Flavour

In a research testing different types of food on different kinds of material cutlery, the wooden cutlery scored significantly lower than stainless steel and plastic cutlery, due to the atypical flavor. The ‘taste of the meal’ also scored lower with the wooden cutlery than with the other materials. The participants said that the ‘saltiness’ and ‘flavor of spices’ were suppressed in contrast with the stainless steel and plastic cutlery. The participants commented that it was because their tongue came into contact with the wooden spoon and that the spoon had an atypical flavour and not that the food was tainted from the wooden spoon. That means that the wooden material did not transfer the taste to the food. [25]

4.4 Findings

The main problems regarding wooden single-use cutlery found in the literature are an atypical woody flavour and suppressed food flavour, high roughness and low shore hardness. Yet, other variables such as colour, shape and weight can affect the overall liking, but no prescriptive information could be obtained for these variables.

A higher roughness in the dining environment is associated with relaxed and warm dining environments, while smooth materials are associated with cleanliness and modernity. Another association made in this research linked a higher shore hardness to a more functional, formal and reliable setting, while a lower shore hardness was found to be related to a casual and relaxed setting. Another association with shore hardness was also found, claiming that a higher shore hardness would enhance taste associations.

Although links exist between colour and taste, no significant effects were found regarding the overall liking of food when taken from differently coloured spoons. Spoon shape can affect the perception of yoghurt, but no prescriptive data can be gathered from the research.

5 Possible Solutions

The main problems that were found in the problem definition were: an atypical taste, surface roughness was too high, and low hardness. For these problems, several options were explored to address them, specifically surface treatments, coatings, and textures.

5.1 Current manufacturing

Wooden cutlery is now made by selecting the right materials. Then, the wood is sliced into thin sheets. Next, the thin wooden sheets go into a die-cutting machine, and the mold that is used creates the shape of the cutlery. After the cutlery is die-cut, the edges of the cutlery are shaped so that the splinters and any extra roughness are removed. This is typically done by sanding machines or tumbling the cutlery in a large drum with abrasive materials. For the round forms of the cutlery, heat and pressure are applied to shape the cutlery. [7] A coating is added at the end to ensure that the cutlery is protected against moisture and food-safe [14].

5.2 Surface Treatments

With surface treatments that exist for wood, it may be possible to create a lower surface roughness and a increased shore hardness for wooden cutlery. There is, however, not a lot of research about wooden cutlery, so the actual surface roughness and hardness cannot be found in articles.

5.2.1 CNC

An experiment was done on the machining of wood surfaces. In the experiment, they used a computer numerical control (CNC). They experimented with the milling cutter trajectory, milling cutter speed and along and against the grain. [9]

For the milling cutter speed, they tested three different speeds: 9.000r/min, 15.000r/min and 18.000r/min. They observed the results with a 100x magnification electron microscope. It became clear that with a spindle speed of 18.000r/min the surface showed a few surface burrs; however, with this spindle speed the surface was smoother than with the other spindle speeds. With the spindle speed of 9.000r/min, the surface has a lot more burrs that are also elongated and you can see the gear marks in the surface. [9]

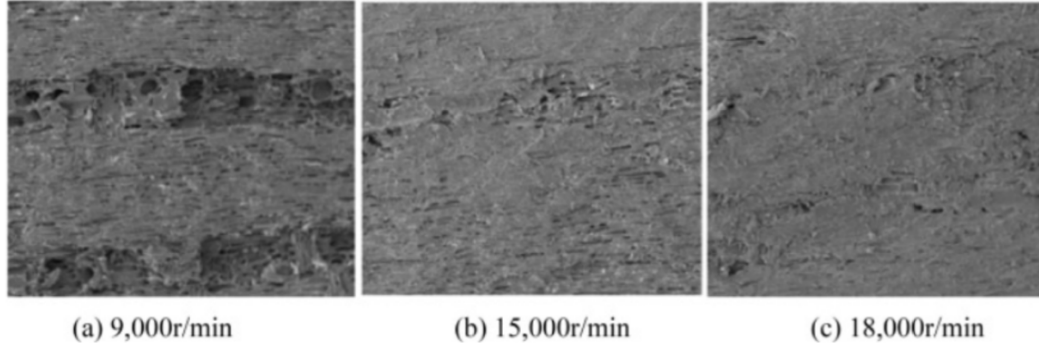


Figure 3: Milling Speeds

They also tested the effect of going against and along the grain and observed with the same electron microscope. The spindle speed of the mill was set to 9.000r/min. When the mill went against the grain, the surface was smoother than when the mill went along the grain. There were no large burrs visible, while the against-the-grain had flaky burrs. [9]

Another research on wood machining experimented on the effects of changing the feed rate, spindle speed, step-over and tool diameter. They used three methods of milling for the experiment: conventional milling offset (up milling), climb milling offset (down milling) and raster. For the experiment, there were 3 states in each state, one parameter would change. In state I, the feed rate, in state II, the spindle speed and in state III, the step-over were the factors that changed. Each state was tested with three different diameter tools: 3, 4 and 6 mm tools. [32]

From the experiment, it became clear that for the three methods of milling, raster milling had the best values for the surface roughness. The tool diameter has a significant effect on the surface roughness. The tool diameter of 6 mm has a higher surface roughness than the tool diameter of 3 mm. For the feed rate, a higher feed rate would result in a higher surface roughness. Lastly, the spindle speed in this experiment went from 15.000r/min to 25.000r/min, which is even higher than the other experiment. For this experiment, it became clear that the surface roughness was better at a higher spindle speed. [32]

5.2.2 High-speed friction

Another research focused on the high-speed friction (HSF) method, which is a new wood surface treatment technology. In this surface treatment technology, the wood surface is rubbed onto a smooth metal surface at high speeds. In this experiment, they used stainless steel tools; one of the tools was not coated, and the other was coated with polytetrafluoroethylene. For the wood, they used air-dried spruce. [12]

For HSF, the wood was secured and only moves on the X-axis, while a round metal bar rotates at a high speed on it. This creates heat and this softens the surface. An illustration of this can be seen in the figure below. [12]

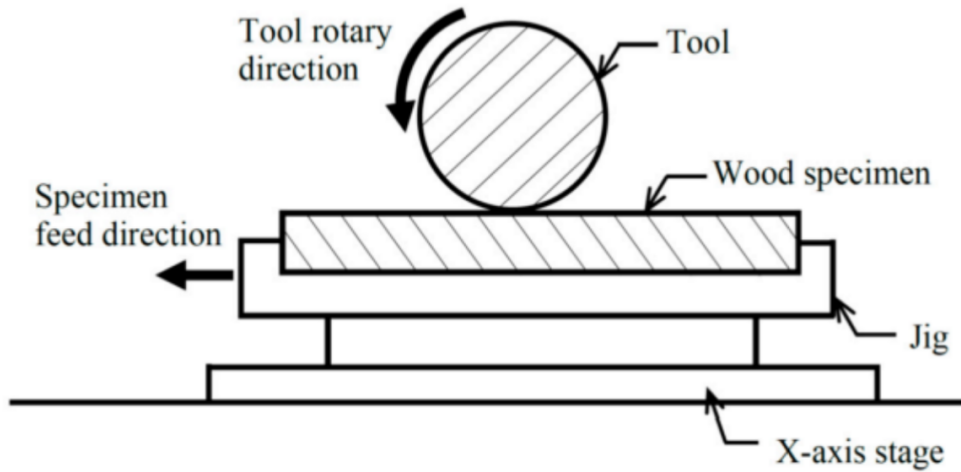


Figure 4: High Speed Friction

After the process the spruce wood sample had a better surface roughness. With the coated tool, the surface roughness was even better. Another benefit of adding the fluorine coating is that the water repellency gets improved.[12]

5.2.3 Thermomechanical densification

In the process of thermomechanical densification, heat and pressure are simultaneously applied to the wood. This changes the material properties so that the cellular structure gets denser. This improves the surfaces of the wood to be smoother. [5] Not only does this create a smoother surface, but it can also improve the hardness. The experiment that was done on improving the hardness had a control sample with a hardness of 2.22kN and the highest hardness after thermomechanical densification was 7.90kN, which is about 3.5 times higher. This was achieved after 10 minutes of densification at 160 °C. [3]

In this research, they experimented with different temperatures and compaction rates. The temperatures they used are 140 °C, 160 °C and 180 °C. The compaction rates they used are 20 % and 40 %. They applied this for 30 minutes. The press they used for this was a hydraulic press with a pressure of 2.5 MPa. [5]

It became clear that the lowest temperature, 140 °C, with a compaction rate of 40% had the best surface roughness. The highest temperature, 180 °C, with a compaction rate of 20% had the worst surface roughness rate. [5]

5.3 Ultrasonic densification

Ultrasound is acoustic energy, and it is widely used in engineering. In wood, there are water particles in the tiny spaces and channels. Ultrasound can make the wood denser with the help of this water. The ultrasound breaks the chemical bonds in the wood, increases how easily the liquids move through the wood, and creates reactive particles. These factors help the wood be compressed and keep the shape during densification. [6] Ultrasonic processing of wood can increase the hardness of the treated surface. [30]

5.3.1 Findings

Several surface treatments can be used to have a better surface roughness. For computer numerical control, different factors were changed to find out what influence these factors have on the surface roughness. It became clear that a high spindle speed and a low feed rate would improve the surface roughness. Also, going along the grain and having a smaller diameter tool has a significant impact on the surface roughness. High-speed friction is another process to improve the surface roughness of wood. In the research, it became clear that if you add fluorine to the metal tool during this process, the roughness of the surface becomes even smoother. Thermomechanical densification changes the material properties of the wood, which results in a smoother surface. For this process, the preferred temperature was around 140 degrees and a compaction rate of 40%. Lastly, ultrasonic densification could also increase the surface hardness.

5.4 Coatings

5.4.1 Edible Coatings

Edible coatings are now commonly used in the preservation of fruits and vegetables. [18] They even use this type of coating on meat and dairy products. [31] This means that they are safe to eat, which is also necessary for the wooden cutlery, since that comes into contact with your mouth. These types of coatings are usually not removed when consumed. [18] However, the coating can be washed away before eating. [17] It usually also does not influence taste, texture and/or appearance. These coatings however should still be safe for humans when consuming food. For edible coatings different bio-based materials can be used, such as protein, lipid and polysaccharides. [18]

5.4.2 Film Coating

Film coating is used for taste masking of drugs for patients. The medical drugs often have a bitter taste which makes the experience for the patient worse. A film coating is often achieved by using different kinds of polymers. The polymers that are used are: water-soluble, cationic, anionic or neutral insoluble. These polymers can all be applied for taste masking. [16]

5.4.3 Findings

There are a few coatings that can be used to mask the atypical flavor of the wooden cutlery. First, there is the edible coating, which is used for fruits and vegetables. Which means it is safe to consume. The coatings are made to be odorless, transparent and have no influence on taste. However, the coating can be washed away, which makes it a less suitable coating in this case. Then the second coating type is a film coating. This type of coating is used for medical drugs to mask the bitter taste of the drugs. The film consists of polymers that can mask the taste of the drug.

5.5 Textures

Although textures are widely acknowledged to influence flavour in foods, no research was found on the influence of utensil textures on the consumption of said food. One study considered the texture of food packaging, finding a significant difference in perceived crunchiness

and hardness of biscuits when taken from a rough pot compared to a smooth pot. Perceived freshness and overall liking of the biscuits did not significantly differ between the pots, nor did the varying pots influence perceived creaminess, thickness, granularity or roughness of yoghurt significantly. [26] Another study found a logarithmic proportional relationship between the viscosity of oral stimulation and the taste cortex, a part of the brain that responds to flavour and its intensity. [2]

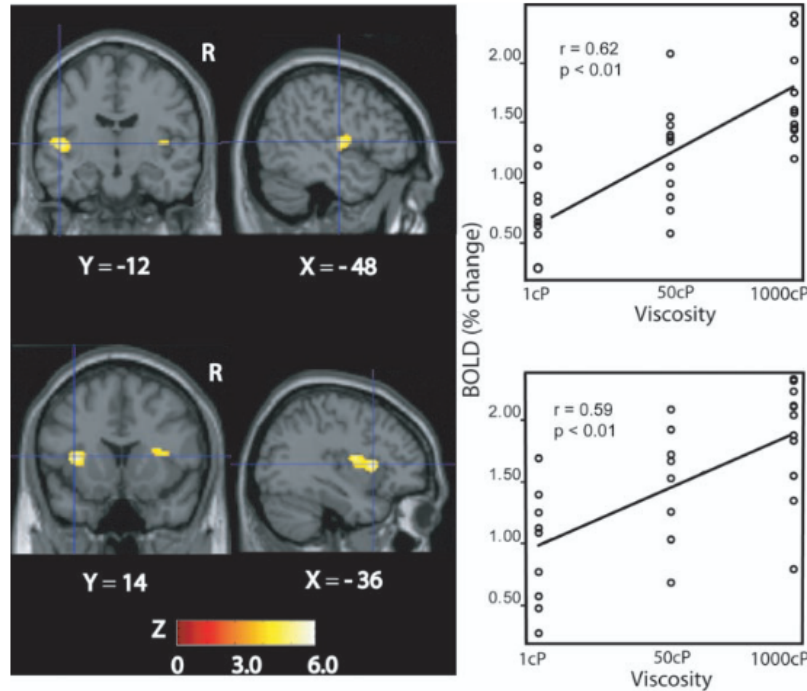


Figure 5: Taste cortex response to viscosity [2]

A group of students created a spoon that would trigger cell-surface receptor proteins that in this case typically react to sweeteners and sugar. The spoon has bumps on the bottom of the spoon and are coated with a layer of molecules. The molecules are called ligands, and these molecules trigger the cell receptors. The spoon does not add any calories or chemicals, but it makes the food taste sweeter. [4]

One research tested the perceived friction, moistness, roughness, stiffness and warmth when altering material texture and strain. Adding texture to increase perceived hardness is researched; an increase in strain showed a higher perceived hardness than the strain actually changed. The article mentions using stiffness instead of hardness and uses the shore scale; it does not mention why this was done. When adding grooves, the perceived hardness is much higher; however, perceived roughness also increases with groove width. [19]

In general, the added texture was perceived when the groove widths changed from 37 to 45 μm . Adding grooves with a minor width to decrease perceived roughness while increasing perceived stiffness could aid in the perception of single-use wooden cutlery.

Making these grooves between 10 and 30 μm has significant changes in perceived stiffness

but not yet in perceived roughness or friction, as can be seen in Figure 6.

The effect of strain and textures has not yet been researched on the oral tactile experience, and the material used in the research is silicone instead of wood; the effect of these changed characteristics will have to be investigated further before being applied to single-use wooden cutlery.

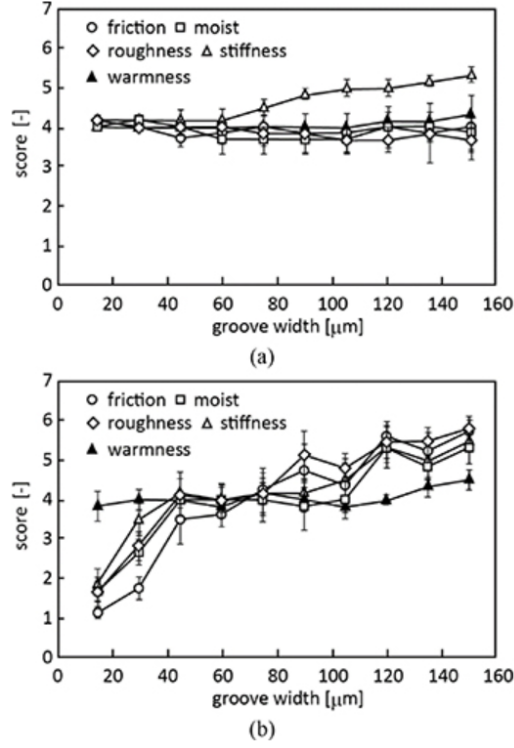


Figure 6: a) samples without grooves, and b) for tactile samples with grooves [19]

5.6 Findings

Several surface treatments can be used to create a better surface roughness. For CNC, different factors were changed to find out what influence these factors have on the surface roughness. It became clear that a high spindle speed and a low feed rate would improve the surface roughness. Also, going along the grain and having a smaller diameter tool have a significant impact on the surface roughness. High-speed friction is another process to improve the surface roughness of wood. In the research, it became clear that if you add fluorine to the metal tool during this process, the roughness of the surface becomes even smoother. Thermomechanical densification changes the material properties of the wood, which results in a smoother surface and an increase in hardness. For the process of creating the smoother surface, the preferred temperature was around 140 degrees and a compaction rate of 40%. Lastly, ultrasonic densification could also increase the surface hardness.

There are a few coatings that can be used to mask the atypical flavor of the wooden cutlery. First, there is the edible coating, which is used for fruits and vegetables. Which

means it is safe to consume. The coatings are made to be odorless, transparent and have no influence on taste. However, the coating can be washed away, which makes it a less suitable coating in this case. Then the second coating type is a film coating. This type of coating is used for medical drugs to mask the bitter taste of the drugs. The film consists of polymers that can mask the taste of the drug.

Textures in the dining environment can impact perceived flavours, smooth packaging can make biscuits taken from it be experienced as less crunchy and hard, and the taste cortex activation inside the brain has been observed to respond to the viscosity of food itself. Perceived hardness and roughness can be altered through strain and texture, allowing for better perceived hardness with little perceived roughness changes. This research has, however, not yet been performed on the oral tactile experience, nor when taking wood as the test material.

6 Test plan

The test plan is divided into two parts: an objective part measuring the effectiveness of the solutions on the mechanical and physical parameters found to be relevant to the look and feel of single-use wooden cutlery, and a subjective part on the effectiveness of these solutions on the overall user experience of the cutlery.

6.1 Objective tests

The effect of machining on the hardness is tested using a Shore Durometer, tested on a Shore D scale (for medium-hard to hard materials). The best result will be taken to the second machining step, where the aim is to decrease roughness.

The effect of three different types of machining on the roughness is measured using an atomic force microscope (AFM). The roughness will be characterised using the arithmetic average roughness (Ra).

The best results will be taken to the subjective testing.

6.2 Subjective tests

Tests will be conducted with participants individually, to avoid them influencing each other's opinions. The participants are asked to sit at a table, and will then be blindfolded. In a randomised order to limit bias, they are asked to eat from different bowls containing the exact same yoghurt. The participants will not know they are testing the spoons, and will be asked their opinion on the yoghurt. The following set of spoons will be prepared for the participants;

Sample	Variable
A (Control)	Untreated
B	Machined
C	Textured
D	Machined and Textured

The participants are asked to rate three qualities of the yoghurt on a 7-point Likert scale, these are the smoothness, flavour and overall liking of the yoghurt.

A repeated measures two-way ANOVA statistical analysis is performed on the results of the subjective testing, to see if the null-hypothesis can be rejected for either the machining, texturing or both with significance.

H0a: Machining does not affect the perception of the yoghurt.

H0ab: Texturing does not affect the perception of the yoghurt.

7 Conclusion

In conclusion, improving the look and feel of the cutlery can be done with surface treatments, coatings and textures. From literature research it became clear that there were a couple of problems with wooden cutlery. The main problems that were found were the atypical flavour, high surface roughness and low hardness. The problems found could be categorized into three different categories: tactile, aesthetic and flavour. These problems were found in different experiments that were done, the participants of the experiments needed to taste different kinds of food with different kinds of materials.

For these problems possible solutions were found. For the surface roughness and hardness, a surface treatment would be an option to make the surface smoother and increase the hardness. The surface treatments that have the potential to create a smoother surface roughness are: CNC, High-speed friction and Thermomechanical densification. For increasing the surface hardness, the surface treatments that are possible are: thermomechanical densification and ultrasonic densification.

For the atypical flavour, using a film coating would be the best option. These types of coatings are used for medical drugs to mask the bitter taste.

To create cutlery with a higher perceived hardness, the materials strain and texture can be changed. When the strain is increased the hardness was also perceived higher. Adding textures, in the form of grooves, also increased the perceived hardness, however, this also made the surface roughness higher. So this needs to be taken into consideration.

Finally, a test plan was created to assess these solutions for the issues found in the problem definition, by the use of objective and subjective tests. The surface treatment for the surface roughness and hardness will be measured with objective tests. After these tests, the best results will be taken into the subjective tests, where participants will assess whether the issues are solved by testing the taste of yoghurt. From this they will be asked to assess different factors.

8 Discussion

One of the difficulties encountered includes a limitation of time, as only five credits of study time are accredited to the research.

This research is limited by existing literature, which is sparse in some aspects. In recent years, more light has been shed on the importance of the context of food and the textures of food, but little research has been done on wooden cutlery and its textures.

Another issue prevalent throughout this research was the lack of singled-out variables, making it impossible to take prescriptive learnings. Changing the material will have an effect on texture, aesthetics and weight, for example, making it difficult to find the importance of these variables on their own.

Some issues found did not yet have the required research as to what should be altered, only stating that it might have an effect. These include the shape and colour of cutlery. Therefore, more in-depth research will have to be done on the topic before possible solutions can be proposed.

Other materials are not considered for this research; perhaps other alternatives could provide better dining experiences, given the environmental and user experience constraints.

An unlikely scenario could be that the result will be too good for the user to throw the cutlery away, which might create new issues, such as hygiene.

The literature does not describe how the characteristics of roughness are built up, such as the shape, patterns or local details. Hardness has been written up as stiffness in an important article for this research; the impact on the results measured on participants is unknown.

Another factor that should be considered is the recyclability of the product, as one of the key reasons for introducing wooden counterparts to plastic cutlery is the sustainability of the product. The impact of the extra processing required to increase the user experience on sustainability will have to be determined.

9 Acknowledgements

In the writing process, Grammarly has been used for rephrasing and grammar checking. Google Scholar Labs and Consensus have been used to discover research.

References

- [1] *Household Disposable Wooden Cutlery Market Size, Share, Growth, and Industry Analysis, By Type (Spoon, Fork, Knife, Chopsticks, Others), By Application (Online Sales, Offline Sales), Regional Insights and Forecast to 2035*. <https://www.360marketupdates.com/market-reports/household-disposable-wooden-cutlery-market-401226>. 2026.
- [2] Ivan E. de Araujo and Edmund T. Rolls. “Representation in the Human Brain of Food Texture and Oral Fat”. In: *The Journal of Neuroscience* 24 (2004), pp. 3086–3093. DOI: 10.1523/JNEUROSCI.0130-04.2004. URL: <https://doi.org/10.1523/jneurosci.0130-04.2004>.
- [3] Poorya Kooche Baghy and Stavros Avramidis. “Thermo-hydro-mechanical densification of hemlock”. In: *International Wood Products Journal* 12 (2021), pp. 231–239. DOI: 10.1080/20426445.2021.1957616. URL: <https://doi.org/10.1080/20426445.2021.1957616>.
- [4] Saugat Bolakhe. *Flavor-enhancing spoons and Chopsticks could make food taste better*. May 2024. URL: <https://www.scientificamerican.com/article/flavor-enhancing-spoons-and-chopsticks-could-make-food-taste-better/>.
- [5] Douglas Edson Carvalho et al. “Surface changes in wood submitted to thermomechanical densification”. In: *Maderas-Cienc Tecnol* 26 (May 2024). DOI: 10.22320/s0718221x/2024.42.
- [6] V Chayeuski. *Densification of Wood – Chemical and Structural Changes Due to Ultrasonic and Mechanical Treatment*. https://www.academia.edu/101191200/{D}ensification_of_wood__{C}hemical_and_structural_changes_due_to_ultrasonic_and_mechanical_treatment. [Accessed 15-04-2026]. 2021.
- [7] EcoAdmin. *How are disposable wooden spoons made? A step-by-step guide*. July 2025. URL: <https://ecowaretech.com/how-are-disposable-wooden-spoons-made-a-step-by-step-guide/>.
- [8] Rebecca Goodrum, Bibiana Bartokova, and Poritosh Roy. “Life cycle assessment of banned single-use plastic products and their alternatives”. In: *Microplastics* 3.4 (Oct. 2024), pp. 614–633. DOI: 10.3390/microplastics3040038.
- [9] Qiang Guo et al. “Study on the application of a new surface burr treatment process”. In: *Alexandria Engineering Journal* 71 (May 2023), pp. 1–11. DOI: 10.1016/j.aej.2023.03.032.
- [10] Siyuan Guo, Bingqian Li, and Yuanyuan Liu. “Exploring Cross-Sensory Perception in Dining Environments: The Role of Tactile Surface Properties on Users’ Visual and Gustatory Experiences”. In: *AHFE International*. Vol. 164. AHFE International, 2025. DOI: 10.54941/ahfe1006061. URL: <https://doi.org/10.54941/ahfe1006061>.
- [11] Vanessa Harrar and Charles Spence. “The taste of cutlery: how the taste of food is affected by the weight, size, shape, and colour of the cutlery used to eat it”. In: *Flavour* 2 (2013). DOI: 10.1186/2044-7248-2-21. URL: <https://doi.org/10.1186/2044-7248-2-21>.
- [12] Ryuichi Iida, Tadashi Ohtani, and Mizuki Yurikusa. “Effect of high-speed friction treatment with fluorine components on the changes in spruce surface properties”. In: *Buildings* 14.11 (Nov. 2024), p. 3513. DOI: 10.3390/buildings14113513.

- [13] Visa Immonen. “Functional Ladles or Ceremonial Cutlery? A Cultural Biography of Prehistoric Wooden Spoons from Finland”. In: *Acta Borealia* 19 (2002), pp. 27–47. DOI: 10.1080/08003830215541. URL: <https://doi.org/10.1080/08003830215541>.
- [14] Jason. *Disposable wooden cutlery production process*. URL: https://m.leafyest.com/blog/Disposable-Wooden-Cutlery-Production-Process_b2163.
- [15] Amanda Jonsson et al. “An evaluation of alternative biodegradable and reusable drinking straws as alternatives to single-use plastic”. In: *Journal of Food Science* 86.7 (June 2021), pp. 3219–3227. DOI: 10.1111/1750-3841.15783.
- [16] S. Joshi and H.-U. Petereit. “Film coatings for taste masking and moisture protection”. In: *International Journal of Pharmaceutics* 457 (2013), pp. 395–406. DOI: 10.1016/j.ijpharm.2013.10.021. URL: <https://doi.org/10.1016/j.ijpharm.2013.10.021>.
- [17] Jian Ju et al. “Application of edible coating with essential oil in food preservation”. In: *Critical Reviews in Food Science and Nutrition* 59.15 (Dec. 2018), pp. 2467–2480. DOI: 10.1080/10408398.2018.1456402.
- [18] Arun Karnwal et al. “Natural biopolymers in edible coatings: Applications in food preservation”. In: *Food Chemistry: X* 25 (Jan. 2025), p. 102171. DOI: 10.1016/j.fochx.2025.102171.
- [19] Makoto Kawazoe and Norihisa Miki. “Tactile samples with variable surface textures to investigate tactile perception characteristic”. In: *Journal of Micromechanics and Microengineering* 30 (2020), p. 105011. DOI: 10.1088/1361-6439/ab9f59. URL: <https://doi.org/10.1088/1361-6439/ab9f59>.
- [20] Mahmood A. Khan. “Technological Disruptions in Restaurant Services: Impact of Innovations and Delivery Services”. In: *Journal of Hospitality; Tourism Research* 44 (2020), pp. 715–732. DOI: 10.1177/1096348020908636. URL: <https://doi.org/10.1177/1096348020908636>.
- [21] Robert Kitz et al. “Food packaging during the COVID-19 pandemic: Consumer perceptions”. In: *International Journal of Consumer Studies* 46 (2021), pp. 434–448. DOI: 10.1111/ijcs.12691. URL: <https://doi.org/10.1111/ijcs.12691>.
- [22] Micah Landon-Lane. “Corporate Social Responsibility in marine plastic debris governance”. In: *Marine Pollution Bulletin* 127 (Feb. 2018), pp. 310–319. DOI: 10.1016/j.marpolbul.2017.11.054.
- [23] *New EU rules to reduce, reuse and recycle packaging*. 2024. URL: <https://www.europarl.europa.eu/news/en/press-room/20240419IPR20589/new-eu-rules-to-reduce-reuse-and-recycle-packaging>.
- [24] Jaqueline Sayuri Nishimura and Shirley Gomes Queiroz. “Cultura material E alimentação: A evolução dos utensílios À Mesa e a Consolidação de comportamentos durante A refeição”. In: *DEMETRA: Alimentação, Nutrição; Saúde* 11.4 (Dec. 2016). DOI: 10.12957/demetra.2016.22229.
- [25] Ana Pantović et al. “The sensory and emotional response to different tableware materials”. In: *Foods* 14.18 (Sept. 2025), p. 3151. DOI: 10.3390/foods14183151.
- [26] Yvette Pascua, Hicran Koç, and E. Allen Foegeding. “Food structure: Roles of mechanical properties and oral processing in determining sensory texture of soft materials”. In: *Current Opinion in Colloid Interface Science* 18 (2013), pp. 324–333. DOI: 10.1016/j.cocis.2013.03.009. URL: <https://doi.org/10.1016/j.cocis.2013.03.009>.

- [27] Betina Piqueras-Fiszman and Charles Spence. “Do the material properties of cutlery affect the perception of the food you eat? An exploratory study”. In: *Journal of Sensory Studies* 26.5 (2011), pp. 358–362. DOI: <https://doi.org/10.1111/j.1745-459X.2011.00351.x>. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1745-459X.2011.00351.x>. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1745-459X.2011.00351.x>.
- [28] Charles Spence et al. “On tasty colours and colourful tastes? Assessing, explaining, and utilizing crossmodal correspondences between colours and basic tastes”. In: *Flavour* 4 (2015). DOI: 10.1186/s13411-015-0033-1. URL: <https://doi.org/10.1186/s13411-015-0033-1>.
- [29] D. Tešanović et al. *The influence of the cutlery and dishes on restaurant food taste experience*. Tech. rep. 37. University of Rijeka, Faculty of Tourism and Hospitality Management, Apr. 2016. DOI: none. URL: https://thi.fthm.hr/images/proceedings/2016/THI_April2016_490to501.pdf.
- [30] A. A. Vjuginova et al. “Ultrasonic modification for improvement of wood-Surface properties”. In: *Wood Material Science and Engineering* 18 (2021), pp. 193–200. DOI: 10.1080/17480272.2021.2006778. URL: <https://doi.org/10.1080/17480272.2021.2006778>.
- [31] P.R. Yaashikaa et al. “Recent advances in edible coatings and their application in food packaging”. In: *Food Research International* 173 (Nov. 2023), p. 113366. DOI: 10.1016/j.foodres.2023.113366.
- [32] Saeed Yaghoubi and Farshad Rabiei. “A profound evaluation of different strategies to improve surface roughness of manufactured part in wood-CNC machining process”. In: *Journal of Engineering Research* 13.3 (Sept. 2025), pp. 1854–1860. DOI: 10.1016/j.jer.2024.05.033.